

For questions 1-2, consider the expression  $15y^3 - 10y^2 - 25y$ .

1. Write an equivalent expression that uses the greatest common monomial factor.
  
  
  
  
  
  
  
  
  
  
2. Write an equivalent expression that does not use the greatest common monomial factor.

For questions 3-6, write an equivalent expression for the given expression that uses the greatest common monomial factor.

3.  $3f^3g^2 + 15fg^3 - 27f^2g^2$

4.  $63s^5h^5 + 42s^3h^6 + 35s^4h^2$

5.  $15d^3b^3 - 60db^2$

6.  $12x^7y^6 - 108x^3y^5 - 84x^6y^4$

For questions 7-10, find the greatest common monomial factor of the expression.

7.  $13a^3 - 39a^2b$

8.  $12q^2c^2 + 30qc + 42q^8c^4$

9.  $10r^7m^5 - 55r^2m^3 + 40r^5m$

10.  $18x^5y^3 - 40x^6y^6 - 38xy$

For questions 1-4, consider the trinomial  $x^2 - 5x - 14$ .

1. How many  $+x$  tiles are needed for an algebra tiles diagram of the trinomial?
2. How many  $-x$  tiles are needed for an algebra tiles diagram of the trinomial?
3. Draw a diagram of the algebra tiles that show the trinomial  $x^2 - 5x - 14$ .
4. Complete the table for  $x^2 - 5x - 14$ .

|       |       |       |
|-------|-------|-------|
|       | $x$   | _____ |
| $x$   | $x^2$ | _____ |
| _____ | _____ | $-14$ |

For questions 5-8, consider the trinomial  $x^2 + 8x - 9$ .

5. How many  $+x$  tiles are needed for an algebra tiles diagram of the trinomial?
6. How many  $-x$  tiles are needed for an algebra tiles diagram of the trinomial?
7. Draw a diagram of the algebra tiles that show the trinomial  $x^2 + 8x - 9$ .

8. Complete the table for  $x^2 + 8x - 9$ .

|       | $x$   | _____ |
|-------|-------|-------|
| $x$   | $x^2$ | _____ |
| _____ | _____ | -9    |

9. Complete the table for  $x^2 + x - 30$ .

|       | $x$   | _____ |
|-------|-------|-------|
| $x$   | $x^2$ | _____ |
| _____ | _____ | -30   |

10. Complete the table for  $x^2 - 11x + 28$ .

|       | $x$   | _____ |
|-------|-------|-------|
| $x$   | $x^2$ | _____ |
| _____ | _____ | 28    |

For questions 1-5, factor each quadratic expression using the method of your choice.

1.  $b^2 - 2b - 63$

2.  $h^2 + 4h - 77$

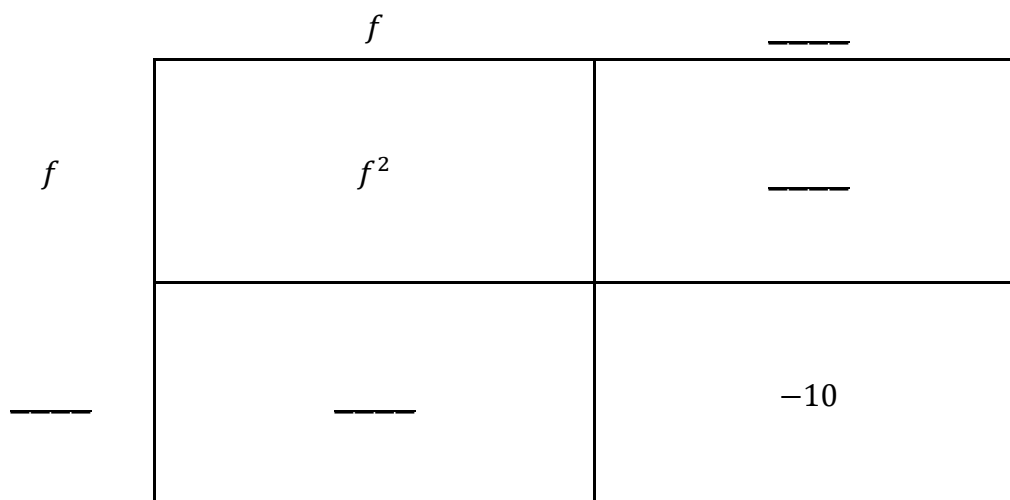
3.  $x^2 - 5x - 50$

4.  $y^2 - 13y + 36$

5.  $x^2 - x - 132$

For questions 6-7, consider the trinomial  $f^2 - 3f - 10$ .

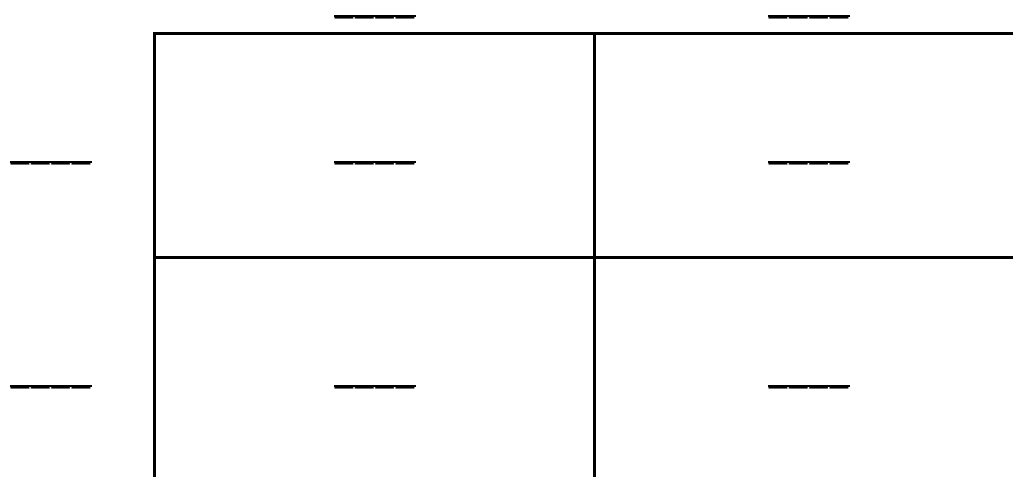
6. Complete the area model.



7. What are the two binomial factors for  $f^2 - 3f - 10$ ?

For questions 8-9, consider the trinomial  $m^2 - 22m + 120$ .

8. Complete the area model.



9. What are the two binomial factors for  $m^2 - 22m + 120$ ?

10. Factor  $x^2 - x - 20$  using factoring by grouping.

For questions 1-5, factor each quadratic expression using the method of your choice.

1.  $5b^2 + 25b + 20$

2.  $4a^2 - 13a - 35$

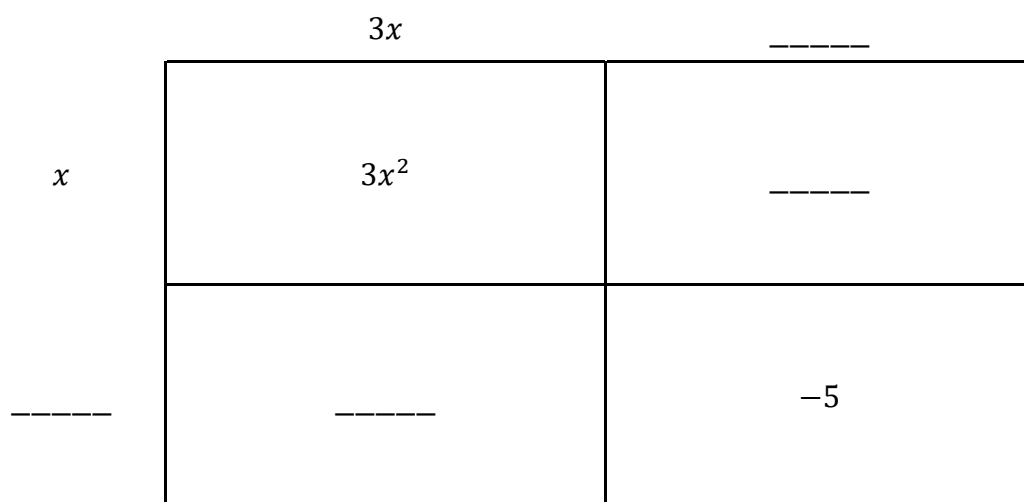
3.  $2x^2 + 5x - 33$

4.  $3k^2 - 27k + 60$

5.  $2z^2 + 13z - 70$

For questions 6-7, consider the trinomial  $3x^2 - 2x - 5$ .

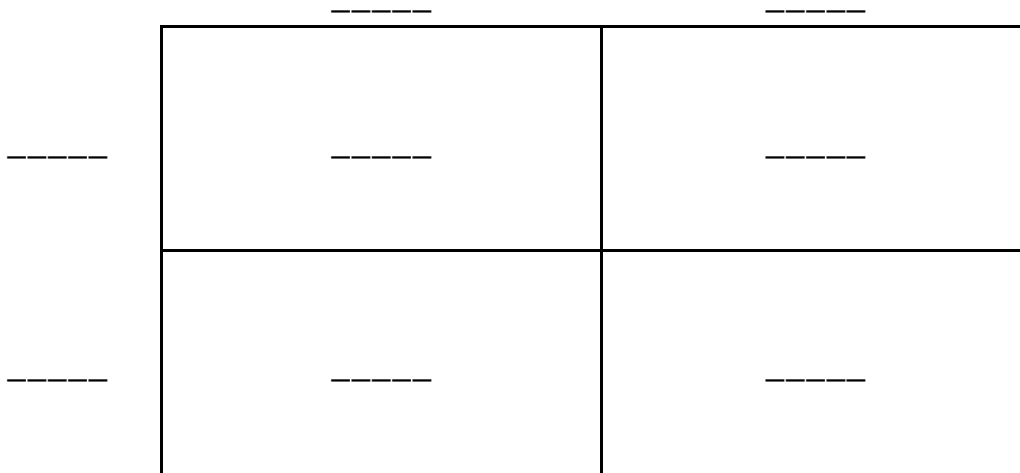
6. Complete the area model.



7. What are the two binomial factors for  $3x^2 - 2x - 5$ ?

For questions 8-9, consider the trinomial  $2m^2 + 11m - 63$ .

8. Complete the area model.



9. What are the two binomial factors for  $2m^2 + 11m - 63$ ?

10. Factor  $3x^2 + 22x + 24$  using factoring by grouping.



For questions 1-3, factor each quadratic expression using the method of your choice.

1.  $4b^2 + 8b - 45$

2.  $12x^2 + 7x - 49$

3.  $10d^2 - 9d + 2$

For questions 4-5, consider the trinomial  $9x^2 - 27x + 14$ .

4. Write an expanded form of the trinomial that can be used to factor the expression by grouping.

5. Factor the expression by grouping.

For questions 6-7, consider the trinomial  $6x^2 - 7x - 20$ .

6. Complete the area model.

|       |      |        |       |
|-------|------|--------|-------|
|       |      | $3x$   |       |
|       | $2x$ | $6x^2$ | _____ |
|       |      |        | _____ |
| _____ |      | _____  | $-20$ |

7. What are the two binomial factors for  $6x^2 - 7x - 20$ ?

For questions 8-9, consider the trinomial  $8t^2 + 18t - 5$ .

8. Complete the area model.

|       |       |       |
|-------|-------|-------|
|       |       | ----- |
| ----- | ----- | ----- |
| ----- | ----- | ----- |

9. What are the two binomial factors for  $8t^2 + 18t - 5$ ?

10. Factor  $10x^2 - 3x - 1$  using factoring by grouping.

Algebra 1

Unit 7

Lesson 6 Homework

Name \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

For questions 1-6, factor each expression by grouping.

1.  $2y^4 - 6y^3 + 14y^2 - 42y$

2.  $15x^2 + 34x + 15$

3.  $-6a + 24ab - 20b + 5$

4.  $6x^2 - x - 40$

5.  $3x^2 + 5x + 6x + 10$

6.  $5x^2 + 52x + 63$

For questions 7-10, consider the trinomial  $16x^2 + 16x - 5$ .

7. List all the factor pairs for  $-80$ .
  
  
  
  
  
  
  
  
  
  
8. Find the factor pair for  $-80$  that add to the sum of  $16$ .
  
  
  
  
  
  
  
  
  
  
9. Use question 8 to write the expanded form of  $16x^2 + 16x - 5$  that can be used to factor the expression by grouping.
  
  
  
  
  
  
  
  
  
  
10. Factor the expression  $16x^2 + 16x - 5$  by grouping.

Algebra 1

Unit 7

Lesson 7 Homework

Name\_\_\_\_\_

Date\_\_\_\_\_

Period\_\_\_\_\_

For questions 1-2, consider the perfect square trinomial  $9x^2 + 24x + 16$ .

1. Rewrite the trinomial in the form  $a^2 + 2ab + b^2$ .

2. Factor the perfect square trinomial.

For questions 3-4, consider the perfect square trinomial  $16x^2 - 40x + 25$ .

3. Rewrite the trinomial in the form  $a^2 - 2ab + b^2$ .

4. Factor the perfect square trinomial.

For questions 5-8, determine whether the quadratic expression is a perfect square trinomial. Then, factor the expression.

5.  $w^2 - 20w - 100$

6.  $f^2 - 2fg + g^2$

7.  $25v^2 - 9$

8.  $9y^2 - 81y + 18$

For questions 9-10, factor the trinomial.

9.  $25a^2 + 120a + 144$

10.  $p^2 - 22p + 121$

Algebra 1

Unit 7

Lesson 8 Homework

Name \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

For questions 1-2, consider the expression  $16m^2 - 100p^2$ .

1. Rewrite the expression as a difference of two squares.

2. Factor the expression.

For questions 3-4, consider the expression  $144a^2 - 49b^2$ .

3. Rewrite the expression as a difference of two squares.

4. Factor the expression.

For questions 5-8, determine whether the quadratic expression is a difference of two squares. Then, factor the expression.

5.  $64d^2 - 121g^2$

6.  $9f^2 + 48f + 64$

7.  $1 - r^2$

8.  $-121 + t^2$

For questions 9-10, factor the expression.

9.  $x^2 - 2x + 1$

10.  $16y^2 - 25x^2$